

Fully Printed Silver-Nanoparticle-Based Strain Gauges with Record High Sensitivity

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This paper describes a strategy to impart brittle conductive patterns composed of silver nanoparticles with high stretchability and structure-dependent electrical characteristics. Silver nanoinks are printed on an elastomeric polyurethane acrylate substrate in the form of planar serpentine structures that can effectively mitigate strain concentration. The relative changes in resistance ($\Delta R/R_0$) and stretchability are found to strongly depend on the serpentine radius (r) that determines the strain relieving efficiency. Features with small radius of curvature show colossal $\Delta R/R_0$ and hold great promise as ultrasensitive stretchable strain gauges. A record high gauge factor of 10^7 is achieved at 12% strain with $r = 200 \mu\text{m}$. Devices with larger radius of curvature exhibit higher stretchability and much more stable conductance, thus can be used as stretchable conductors. The results demonstrate the versatile functionalities that can be acquired from conventional materials by judicious structural designs.

Strain sensors, also known as strain gauges, with the capability of transducing tensile or compressive strains to electrical signals that are usually in forms of change in resistance or capacitance, are widely used in measuring strain, tension, and acceleration, as well as structural damage diagnosis. Conventional strain gauges are made of metal foils or semiconductor films and register changes in resistance upon dimensional changes of the material. The sensitivity of a resistive strain gauge, also termed as gauge factor, is defined as the rate of relative change in electrical resistance to mechanical strain. Metal foil strain gauges usually possess a gauge factor of ≈ 2 and a detection range of $< 5\%$. Semiconductor-based strain gauges have a much higher gauge factor (≈ 200) but with a very limited detection range. In recent years, researchers have made significant progress in improving the sensitivity and detection limits of strain gauges by taking advantage of new nanomaterials and

nanotechnologies.^[1] In particular, although structural crack formation is usually undesired and deemed as an indication of catastrophic structural failure, it can be a very effective approach for achieving ultrahigh gauge factors if the nanoscale cracks are formed in a controlled manner. In fact, crack-enabled ultrasensitive mechanodetection can be readily found in nature, like the crack-structured slit organs near the leg joints of spiders that impart them with extreme sensitivity for tiny mechanosignals.^[2] Very recently, nanometer-scale crack-enabled ultrasensitive strain gauges have been reported with a gauge factor of 2000 and excellent performance in vibration detection.^[3] However, the detecting range is limited to 2%. In fact, a large detection range ($> 5\%$) is usually required for novel applications like wearable elec-

tronics and humanoid robotics.^[4] During the last several years, numerous stretchable strain gauges that can detect strains up to 100% have been reported.^[4b,c,5] Nonetheless, most of them show limited gauge factors (< 10). Furthermore, strain gauges that are fabricated by printing have been very rarely studied.^[6]

Printing as a new approach to fabrication of electronic devices have attracted extensive research efforts due to its unique advantages in reducing manufacturing cost, diminishing hazardous waste, enlarging device size, and liberating the devices from rigid substrates.^[7] Literally any material that can be dissolved or dispersed in a liquid phase, like small molecules, polymers, nanomaterials, and even cells, could be processed by printing.^[8] Representative nanomaterials under current explorations include metallic nanoparticles, semiconducting nanocrystals, nanotubes/nanowires, and, more recently, 2D nanoflakes.^[9] All these nanomaterials can be printed on flexible substrates, yet only nanotubes/nanowires have been endowed with stretchability that is essential for building genuinely mechanically robust electronic systems.^[10] The mechanism underlining the stretchability of nanotubes/nanowires is that their networks can accommodate a certain amount of tensile strain by mesh deformation while still maintaining good percolating pathways for electrical conduction. The 1D nanomaterials themselves in such networks only undergo insignificant amount of stretching. On the contrary, for nanoparticles and nanocrystals, such network structure is absent, hence they are usually very brittle and undergo catastrophic failure even under very small tensile strains.

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There are two primary strategies to make stretchable features out of brittle materials.^[11] The first approach is to disperse the active material in an elastomeric matrix.^[12] Such composites are simultaneously conductive and stretchable when the nanofillers form percolating pathways. Actually, this method has been used for decades in making conductive rubbers with carbon black particles as nanofillers. A high loading of nanoparticles is usually needed to obtain a decent electrical performance due to their low aspect ratio. Yet, the best attainable performance can be orders of magnitude lower than that of pure nanoparticles because the polymer molecules surrounding the nanoparticles act as conduction barriers that the carriers must overcome through tunneling. The alternative strategy is to exploit new strain absorbing structures, like in-plane wavy and serpentine structures as well as out-of-plane buckling structure where a large fraction of the external strain is accommodated by the rotation of the segments instead of being experienced directly by the thin-films or filaments.^[13] Thanks to the progress in nanomembrane preparation and deep insights into the mechanics of stiff-film-on-compliant-substrate systems, numerous single crystal metals and inorganic semiconductors have been made highly stretchable while maintaining their superior electrical characteristics.^[14] The fabrication of such devices, however, relies on microfabrication techniques and vacuum facilities that may pose considerable limits on scaling up the device size and reducing the cost.

In principle, printing and strategic structural design can be combined to make stretchable devices out of otherwise brittle materials in a low-cost and scalable fashion, which, to the best of our knowledge, has been rarely explored. Herein, using silver nanoparticles ink as a model 0D nanomaterial, we propose a universal approach to printed nanoparticle-based

stretchable devices with excellent electrical functionality. In this work, serpentine structured silver nanoparticle (AgNP) features with various radii were printed on a polyurethane acrylate (PUA) substrate. Applying tensile strain induces formation of nanoscale cracks in the AgNP features and thus a change in resistance that depends on the number and size of cracks, which can be used to quantify the applied strain. The stretchability and strain-induced resistance change of the AgNP features were found to be strongly dependent on the curvature radius (r). More specifically, AgNP serpentine features with r of 200 μm were found to exhibit gigantic changes in resistance ($\Delta R/R_0$), with a gauge factor (defined as $(\Delta R/R_0)/\epsilon$ where ϵ represents tensile strain) of $>10^7$ at an ϵ of $\approx 12\%$; while those with r of 1600 μm exhibited much better stretchability and more stable conductance, with $\Delta R/R_0$ of ≈ 10 at an ϵ of $\approx 25\%$. Finite element analysis and in situ optical microscopic observation revealed that the serpentine structures could indeed effectively relieve the strain inside the features and larger curvature radii would lead to better strain relieving efficiency. Noticeably, all printed AgNP features possessed very stable resistance-strain relationship under long-term cyclic stretching tests. Depending on their rate of resistance change $(\Delta R/R_0)/\epsilon$, the printed serpentine AgNP features could be used as either ultrasensitive, stretchable strain gauges with a record high gauge factor of $\approx 10^7$ at 12% strain, or electrical interconnects for stretchable integrated circuits. The strategy presented here could be readily expanded to the fabrication of various printed stretchable electronic devices with tunable performance using other intrinsically brittle nanoparticles or nanocrystals.

The schematic of the printing process is illustrated in **Figure 1a**. A PUA substrate was first treated with O_2 plasma or corona to facilitate wetting of the silver nanoink (PG-007AA,

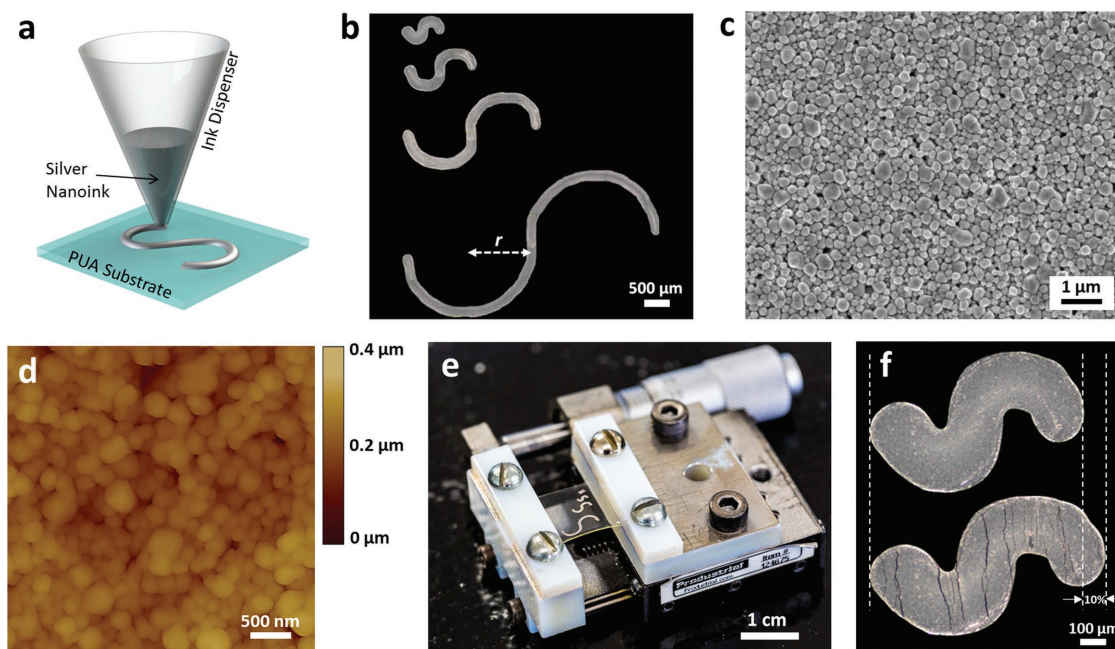


Figure 1. Fully printed AgNP serpentine features on PUA substrate. a) Schematic diagram of the device and the printing process. b) Optical microscope image of the printed devices with radii of 200, 400, 800, and 1600 μm , respectively. c) SEM image of the printed AgNP thin film. d) AFM image of the printed AgNP thin film. e) Photo showing a sample mounted on a linear stage for the stretching test. f) Optical images showing a serpentine feature with a radius of 200 μm when stretched to 10% strain.

Paru, South Korea) on the surface. The AgNPs were then printed onto the treated PUA substrate using a GIX Microplotter (Sonoplot Inc.) with predesigned patterns. Finally, the AgNP features were annealed in air at 120 °C for 1 h to improve their conductivity by removing the polymer binders and sintering the nanoparticles. The radii of the serpentine AgNP features were varied from 200 to 1600 μm while the feature width was kept ≈ 200 μm (Figure 1b). PUA was selected as the elastomeric substrate because, compared to polydimethylsiloxane (PDMS), it has a much higher Young's modulus (≈ 38 MPa vs ≈ 3 MPa) and lower linear coefficient of thermal expansion (≈ 160 ppm $^{\circ}\text{C}^{-1}$ vs ≈ 320 ppm $^{\circ}\text{C}^{-1}$),^[15] thus exhibits a less pronounced mismatch with AgNP features. In fact, the AgNP features printed on the PDMS substrate were not conductive after annealing due to the formation of numerous thermal stress induced cracks. Figure 1c presents the optical image of four AgNP serpentine features with different radii r printed on a single PUA substrate. As shown in the scanning electron microscopy (SEM) and atomic force microscopy (AFM) images in Figure 1c,d, the Ag nanoparticles are around 200–300 nm in diameter.

The electromechanical properties of the printed serpentine AgNP patterns were subsequently characterized in detail. We monitored the change in resistance of all four features while stretching the PUA substrate using a linear stage (Figure 1e). As shown in Figure 1f, lots of microcracks appeared in the AgNP feature with $r = 200$ μm under a tensile strain of 10%. Many of the cracks are straight and almost across the entire feature, indicating the brittle nature of AgNP assemblies. Such crack formation happened to all printed AgNP features, but the location, amount and size of the cracks, as discussed

below, varied significantly depending on the pattern's radius of curvature. These crack formation induced a drastic change in the device's electrical resistance as shown in Figure 2, where a strong dependence of $\Delta R/R_0$ on r can be found. It should be noted that the crack formation during the very first stretching cycle is an irreversible process, thus the resistance will not fully return to the pristine value when the strain is released (details can be found in Figure S1 of the Supporting Information). This is a universal phenomenon observed in many stretchable devices. After the first few stretching cycles, the device performance settles and its resistance–strain characteristics becomes stable. All data presented in Figure 2 are collected from devices whose electrical characteristics have stabilized.

Figure 2a presents the relative change in resistance $\Delta R/R_0$ plotted as a function of ϵ for AgNP features with $r = 200, 400, 800$, and 1600 μm , as well as a straight line ($r = \text{infinity}$). As expected, the straight line can only be stretched to $<2\%$ with a $\Delta R/R_0$ of nearly 12 orders of magnitude, while the serpentine structured patterns are significantly more stretchable. A clear monotonic trend is observed—a larger radius of curvature leads to a better stretchability. Specifically, the sample with $r = 200$ μm could be stretched to 12% before becoming insulating, while the sample with $r = 1600$ μm could be stretched up to 25% with $\Delta R/R_0$ of only ≈ 13 . The stretchability of the $r = 1600$ μm sample was actually limited by the PUA substrate that ruptured beyond 25% strain. It is worth noting that, except for the $r = 1600$ μm sample, all other samples exhibit gigantic $\Delta R/R_0$, being 36, 4.2×10^2 , and 6.6×10^4 at $\approx 10\%$ strain for 800, 400, and 200 μm features, respectively, implying they could be used as ultrasensitive stretchable strain sensors. Figure 2b presents the gauge factors at different tensile strains for all four

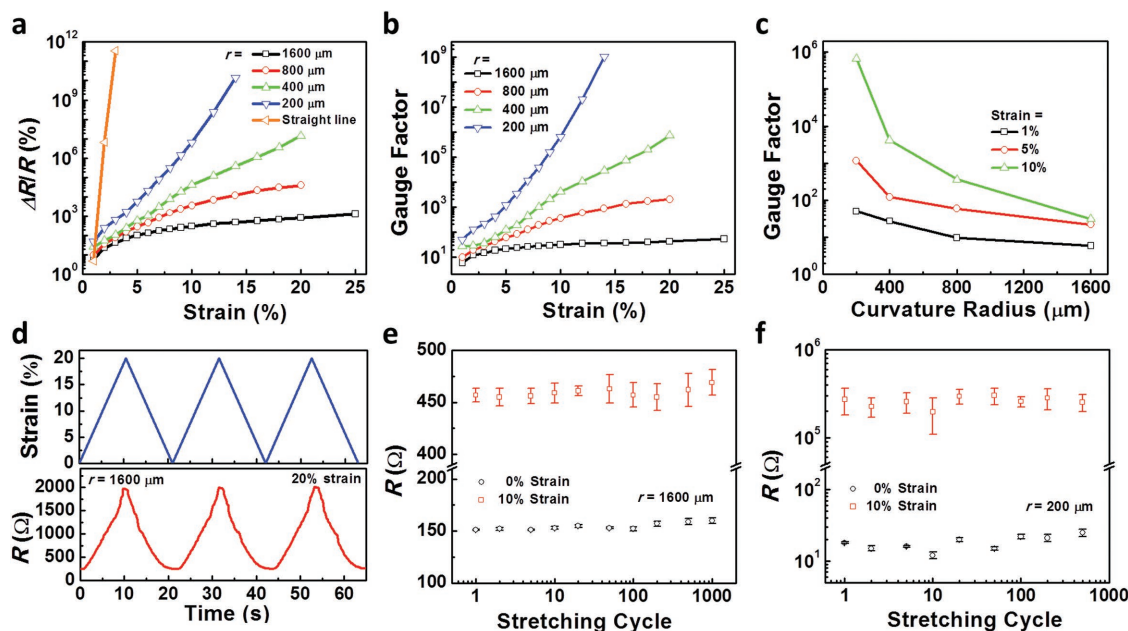


Figure 2. Electromechanical characterization of the printed serpentine features. a) Relative change in resistance plotted as a function of tensile strain for AgNP features with different radii. b) Gauge factors for samples with different radii extracted from panel (a). c) The gauge factor plotted as a function of device's radius. d) Time response of the device with a radius of 1600 μm while being repeatedly stretched to 20% strain and released back to 0% strain. e,f) Long-term stability of the printed serpentine features with a radii of (e) 1600 μm and (f) 200 μm under repeated stretching tests for 1000 cycles.

serpentine features. Notably, the feature with $r = 200 \mu\text{m}$ possesses extremely large gauge factors of $\approx 6.6 \times 10^5$ and 2×10^7 at tensile strains of $\approx 10\%$ and $\approx 12\%$, respectively, marking it the highest record of gauge factor values for stretchable strain sensors ever reported. Our printed AgNP-based strain sensors overwhelmingly outperform conventional metal foil strain gauges in terms of gauge factor (typically ≈ 2) and detection range ($\varepsilon < 5\%$). Figure 2c depicts gauge factors as a function of radius of curvature under different tensile strain values, showing the monotonic decreasing trend as the radius increases, the reason for which will be discussed later in the paper. It is worth noting that, by keeping other parameters like ink formulation, sintering condition, feature width, and film thickness constant, the device performance is primarily determined by only the feature curvature radius and shows decent reproducibility. The data can be found in Figure S2 of the Supporting Information.

We also investigated the long-term stability of the devices during repeated stretch–release tests. The upper and lower panels of Figure 2d show the strain profile and measured time response of resistance for the $r = 1600 \mu\text{m}$ feature, respectively, during three representative stretching cycles. The results show that the measured resistance traces the strain profile nicely, indicating the efficacy of using the device as a strain sensor. Figure 2e,f presents the results of repeatedly stretching the $r = 1600 \mu\text{m}$ and $r = 200 \mu\text{m}$ features to 10% tensile strain for around 1000 cycles. Under 10% strain, the $r = 1600 \mu\text{m}$ feature showed a relatively small resistance change compared to the relaxed state (0%), from ≈ 150 to $\approx 450 \Omega$, while the $r = 200 \mu\text{m}$ feature underwent a drastic resistance change of >4 orders in magnitude. In the meantime, over the period of 1000 stretching cycles, the resistance values in both relaxed and stretched states for both devices exhibit excellent long-term stability, indicating the structural change (the opening and

closing of the microcracks) is reversible and very stable. Furthermore, although significant hysteresis was observed during the first stretching–releasing cycle as discussed previously (Figure S1, Supporting Information), the hysteresis became less pronounced and was eventually negligible after only a few stretching cycles. The hysteresis-free resistance–strain characteristics can be found in Figure S3 of the Supporting Information, which indicates the excellent reliability of the printed AgNP-based strain gauges.

In order to understand the dependence of stretchability on the features' radius of curvature, we carried out finite element analysis (FEA) using COMSOL to investigate the strain distribution throughout the pattern. The FEA results for AgNP features with $r = 200 \mu\text{m}$ and $1600 \mu\text{m}$ are presented in Figure 3. Comparing the strain profiles shown in Figure 3a,b, one can easily draw the conclusion that the larger curvature radius can effectively alleviate the strain directly experienced by the AgNP features—the strain in the $1600 \mu\text{m}$ feature is an order of magnitude lower than that in the $200 \mu\text{m}$ feature. It is also easy to understand this conclusion considering the fact all AgNP features have the same width (w), i.e., a larger radius of curvature means a larger aspect ratio (r/w). Extensive theoretical and experimental studies have revealed that stiff materials can be rendered flexible by increasing the aspect ratio (reducing the thickness in the case of thin films). For instance, $\approx 100 \text{ nm}$ thick single crystalline silicon nanoribbons are highly flexible and experience peak strains of only $\approx 0.0005\%$ when bent to curvature radius of $\approx 1 \text{ cm}$.^[11] Another conclusion that can be drawn from Figure 3 is that the strain is not uniformly distributed throughout the serpentine features. Instead, significant strain concentration occurs, particularly on the inner side of wave crests and troughs (location A in Figure 3a,b), whereas the strains in the vicinity of wave

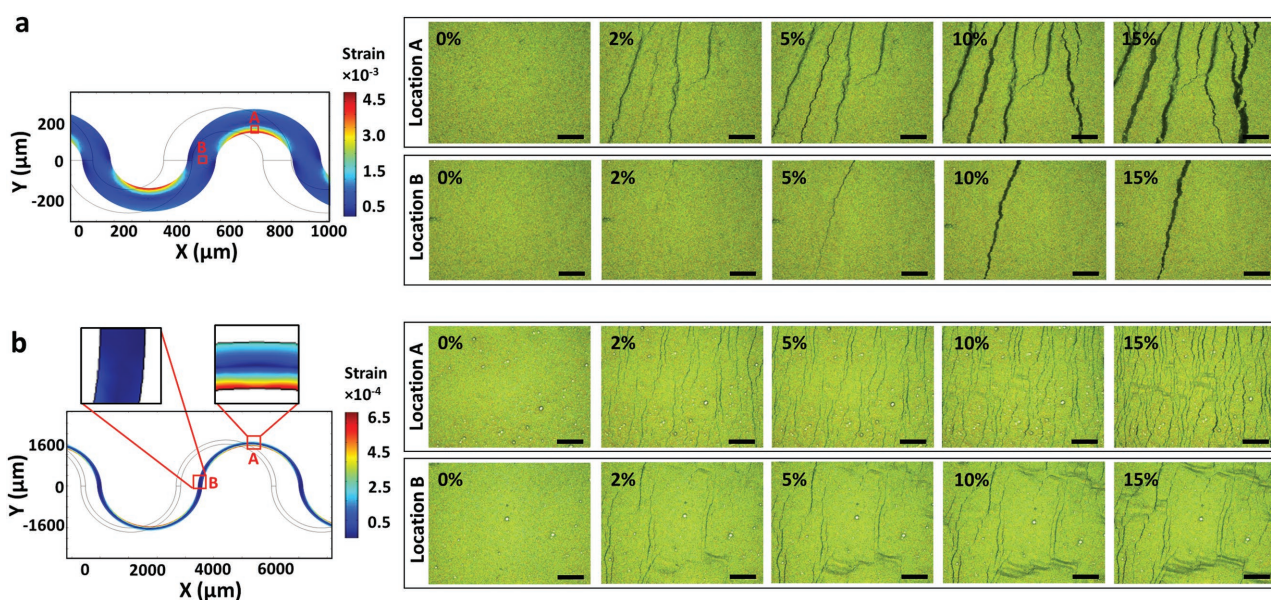


Figure 3. Finite element simulation results and extended depth of field optical images of the printed serpentine features showing the strain distribution and microcrack formation. The left-hand side shows the simulated strain distribution for serpentine pattern with a radius of a) $200 \mu\text{m}$ and b) $1600 \mu\text{m}$ when stretched to 10% strain. The optical images on the right-hand side show the crack formation at locations with the highest strain (Location A) and the lowest strain (Location B) under different level of strain from 0% to 15%. The scale bar in all images represents $10 \mu\text{m}$.

equilibriums (location B in Figure 3a,b) are nearly an order of magnitude lower.

The right-hand side of Figure 3 presents the real-time extended depth of field images at two representative locations (locations A and B) in the 200 μm (Figure 3a) and 1600 μm (Figure 3b) features when the sample is stretched to various tensile strain levels up to 15%. For both features, location A sees a lot more cracks with each crack being wider than location B does, manifesting the obvious strain concentration at location A. Furthermore, by comparing the amount and width of cracks at the corresponding locations in the two features, one can conclude that the $r = 200 \mu\text{m}$ feature experiences more significant strain than the $r = 1600 \mu\text{m}$ feature, which is in good agreement with the FEA results and also explains the monotonic decreasing trend of the gauge factor for the increasing radius of curvature observed in Figure 2.

As discussed above, the printed serpentine features with a smaller radius of curvature exhibit gigantic yet reversible resistance changes under tensile strains of up to 20% and are thus promising platforms for application as ultrasensitive stretchable strain sensors for human motion detection. Here, as a proof-of-concept demonstration, we used a sample with repeated segments of $r = 400 \mu\text{m}$ serpentine features to monitor the bending motion of human figures. We first carried out electromechanical analysis on the sample to calibrate its $\Delta R/R_0$ up to a tensile strain of 16%. The $\Delta R/R_0$ is plotted in Figure 4a as a function of tensile strain, showing a gauge factor of ≈ 700 at 10% strain. The device was then mounted on a nitrile glove that was subsequently worn by a subject. As presented in Figure 4b,c, unambiguous stepwise resistance changes were recorded while the finger was bent gradually. The strain values indicated in Figure 4b were obtained by interpolating the $\Delta R/R_0$ - ϵ curve shown in Figure 4a. Human motion detection is just one example of future applications. Due to the excellent stretchability and record-high sensitivity, we believe our

printed AgNP-based serpentine structures may also see a wide range of applications in flow rate measurements, vibration monitoring, structural damage diagnosis, humanoid robots, and so on.^[3,16]

In summary, we have successfully achieved high stretchability with intrinsically brittle silver nanoparticle printed patterns by taking advantage of planar serpentine structures that, as confirmed by finite element analysis, can effectively alleviate the strain concentration. The device characteristics, in terms of gauge factor and stretchability, can be effectively tuned in a broad range by varying the radius of curvature of the serpentine. Features with the small radius of curvature exhibit gigantic $\Delta R/R_0$ while maintaining excellent stretchability, and are thus promising platforms for ultrasensitive stretchable strain sensors. A record high gauge factor of $>10^7$ and detection range of $\approx 12\%$ are obtained simultaneously in a serpentine feature with $r = 200 \mu\text{m}$. Such strain gauge devices with superior performance can be used for various applications as manifested by the proof-of-concept demonstration of human motion detection. The samples with the larger radius of curvature have higher strain relieving efficiency and result in better stretchability. The feature with $r = 1600 \mu\text{m}$ can be stretched to beyond 25% strain (limited by the rupture of the PUA substrate) with a relatively small $\Delta R/R_0$ of ≈ 10 and works well as stretchable interconnects (see Figure S4 of the Supporting Information). Furthermore, all of our devices are fabricated by printing process that is compatible with low-cost and scalable manufacturing. Future study along this direction is needed to elucidate the influence of AgNP particle size, film thickness, and sintering temperature on the final device performance. This work represents our capability of endowing conventional materials with new attributes and realizing versatile functionalities through judicious structural designs, which could be readily extended to other nanoparticles and nanocrystals.

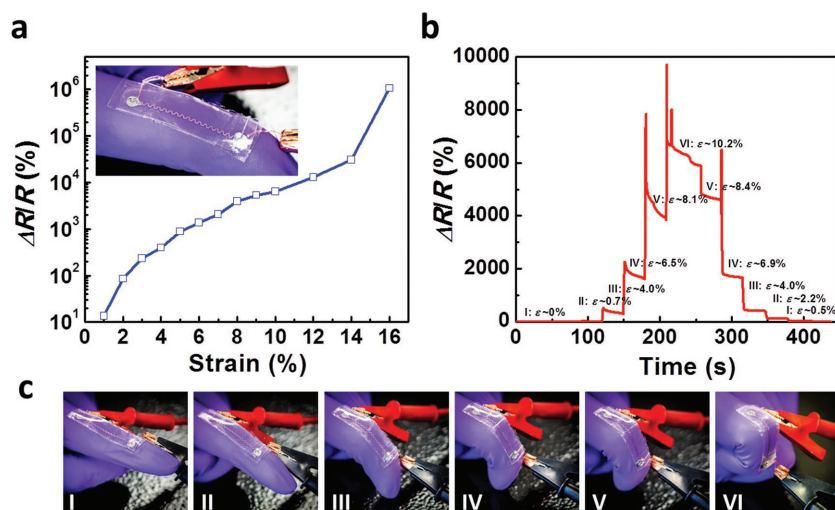


Figure 4. Printed ultrasensitive strain gauge for finger motion detection. a) Relative change in resistance plotted as a function of tensile strain for a sample with 20 repeated segments of $r = 400 \mu\text{m}$ serpentine patterns. Inset: photo showing the sample mounted on a glove used for the finger motion detection. b) Measured electrical resistance from the sensor and deduced strain values for six different states. c) Photos of the six different states when the finger was bent to different angles.

Experimental Section

Preparation for the Substrate: The PUA substrate was made by the mixture of siliconized urethane acrylate oligomer (CN990, Sartomer), ethoxylated bisphenol A dimethacrylate (SR540, Sartomer), and 2,2-dimethoxy-2-phenylacetophenone (photoinitiator, Sigma-Aldrich) with the ratio of 10:1:0.1, followed by UV light curing for 10 min. The CN990 was chosen in order to improve the stretchability of the substrate and the SR540 was chosen to adjust the stiffness of the substrate. A representative stress-strain curve of the PUA substrate is shown in Figure S5 of the Supporting Information.

Device Fabrication: The silver nanoparticle ink (PG-007AA from Paru Corporation, South Korea), dispersed in ethylene glycol with a concentration of 60 wt%, particle size 200–300 nm, was printed onto PUA substrate (pretreated by the oxygen plasma, 60 W for 10 s) using a GIX Microplotter (Sonoplot Inc.). The whole device was then placed on top of a hotplate and annealed for 1 h at 120 $^{\circ}\text{C}$. The annealing condition was chosen such that the PUA substrate remained intact after sintering of AgNPs. For strain gauge application, the liquid metal (LMP-2, ROTOMETALS) and copper wires were put onto the two ends of the devices serving

as the connection for measurement, followed by pouring another layer of PUA solution and then curing by the UV light to form the encapsulation.

Device Characterization: Microscopic morphology of AgNPs film was captured by a Hitachi S-4700II field-emission scanning electron microscope and Dimension 3100 atomic force microscope. The stretching tests were performed by either a linear stretching stage or automatically by a syringe pump (for cyclic stretching tests). The electrical characteristics were measured using an Agilent B1500A semiconductor parameter analyzer. Stress–strain curve of the PUA substrate was collected using a UTS Mechanical Testing System conforming to the ASTM D412 standard.

Finite Element Analysis: The FEA was conducted by using COMSOL Multiphysics 5.2. In the model, serpentine structures with different curvature radii, 1600, 800, 400, and 200 μm were constructed and the triangle mesh was used. During the simulation, one end of the trace is fixed, while a prescribed strain (10% of the length of the trace) was applied to the other end along the length direction. Both stress and strain in the model were calculated and shown by color contour plots.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

printed electronics, silver nanoparticles, strain gauges, stretchable conductors, stretchable electronics

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